

Questions for the Audience

How can we Detect Overfitting via Validation Set?

- Validation loss increases
- Validation accuracy decreases
- Other validation metrics exist, e.g, $\frac{1}{N} \sum_n y_{\tau}^{[n]}$

Which Operations Support Automatic Differentiation?

- Matrix multiplication
- Activation functions
- Convolution (via matrix multiplication)
- Pooling, summing, averaging, indexing
- **Any other differentiable operation**

Questions for the Audience

How many Learnable Parameters does a 3×3 Convolution have?

- $C * Q * 3 * 3 + Q$ for channeled 2D convolution:
 C input channels, Q output channels, each with bias

How does this Reduce when Applying Stride of 2?

- Not at all: stride is on the data, not on the kernel

What Happens when Stacking two 3×3 Convolutions?

- What is the receptive field of this combination?
→ Same as 5×5 convolution, i.e., 5×5 pixels
- Is it strictly necessary to have a non-linearity in between?
→ No. We just get a (reduced) 5×5 convolution
→ It is still advisable to increase non-linearity of network